

ch27 Nuclear Energy

2013 Q36

- *36. The sun releases huge amount of energy through thermonuclear fusion while at the same time its mass decreases. The average power released by the sun is about 3.8×10^{26} W. Estimate the decrease in mass of the sun in one second.

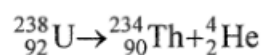
- A. 4.2×10^6 kg
- B. 4.2×10^9 kg
- C. 1.3×10^{15} kg
- D. 1.3×10^{18} kg

2014 Q33

- *33. A radium nucleus decays to a radon nucleus by emitting an α -particle. The energy released in the process is 4.9 MeV. Compared to the mass of a radium nucleus, the total mass of a radon nucleus and an α -particle is

- A. 5.4×10^{-11} kg less.
- B. 5.4×10^{-11} kg more.
- C. 8.7×10^{-30} kg less.
- D. 8.7×10^{-30} kg more.

- *33. The following shows the decay of uranium-238 ($^{238}_{92}\text{U}$).



Given that : mass of $^{238}_{92}\text{U} = 238.05079 \text{ u}$
 mass of $^{234}_{90}\text{Th} = 234.04363 \text{ u}$
 mass of $^4_2\text{He} = 4.00260 \text{ u}$

Which of the following statements is/are correct ?

- (1) The temperature required to start the decay is about 10^7 K .
 - (2) The energy released in the decay of one uranium-238 nucleus is 4.25 MeV.
 - (3) All the energy released in the decay becomes the kinetic energy of ^4_2He .
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only

31. Which of the following nuclear reactions is/are possible for a chain reaction to occur ?

- (1) $^2_1\text{H} + ^3_1\text{H} \rightarrow ^1_0\text{n} + ^4_2\text{He}$
- (2) $^{235}_{92}\text{U} + ^1_0\text{n} \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + 3 ^1_0\text{n}$
- (3) $^{239}_{94}\text{Pu} + ^1_0\text{n} \rightarrow ^{148}_{58}\text{Ce} + ^{89}_{36}\text{Kr} + 3 ^1_0\text{n}$

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

- *33. Given: mass of proton = 1.007276 u
 mass of neutron = 1.008665 u
 mass of ${}^3_2\text{He}$ nucleus = 3.016030 u
 1 u = 931 MeV

When a ${}^3_2\text{He}$ nucleus is formed from 2 protons and 1 neutron,

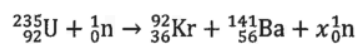
- A. 6.7 MeV energy is released.
- B. 6.7 MeV energy is required.
- C. 8.0 MeV energy is released.
- D. 8.0 MeV energy is required.

- *33. Given: mass of a neutron = 16749×10^{-31} kg
 mass of a proton = 16726×10^{-31} kg
 mass of an electron = 9×10^{-31} kg

In a nuclear reaction, a neutron becomes a proton and a β particle. Estimate the energy released in this process.

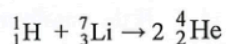
- A. 1.8 MeV
- B. 1.3 MeV
- C. 0.79 MeV
- D. 0.51 MeV

33. What is the value of x for the fission reaction below ?



- A. 1
- B. 2
- C. 3
- D. 4

- *32. When a proton (${}^1_1\text{H}$) bombards a lithium nucleus (${}^7_3\text{Li}$), two α particles (${}^4_2\text{He}$) are produced.



The energy released in this nuclear reaction is 17.32 MeV. Find the mass of a lithium nucleus.

Given: mass of ${}^1_1\text{H} = 1.0078 \text{ u}$

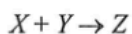
mass of ${}^4_2\text{He} = 4.0026 \text{ u}$

- A. 6.9788 u
- B. 6.9974 u
- C. 7.0017 u
- D. 7.0160 u

33. Which statement about nuclear fission is correct ?

- A. It involves two nuclei combined together.
- B. All nuclear fission reactions are spontaneous.
- C. Rate of fission reaction is temperature dependent.
- D. Fission of heavy nuclei yields products that are more stable in terms of energy.

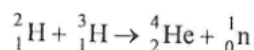
- *33. In the following nuclear reaction, nuclide X and Y fuse to form nuclide Z under suitable conditions and the amount of energy released is 7.16 MeV.



The total mass of X and Y is

- A. 0.00385 u less than the mass of Z .
- B. 0.00385 u more than the mass of Z .
- C. 0.00769 u less than the mass of Z .
- D. 0.00769 u more than the mass of Z .

- *33. Consider the following nuclear reaction:



Using the data below, estimate the energy released when 5 g of ${}^2_1\text{H}$ completely reacts with ${}^3_1\text{H}$ through the reaction.

mass of ${}^2_1\text{H} = 2.0136 \text{ u}$

mass of ${}^3_1\text{H} = 3.0155 \text{ u}$

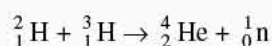
mass of ${}^4_2\text{He} = 4.0015 \text{ u}$

mass of ${}^1_0\text{n} = 1.0087 \text{ u}$

- A. $3 \times 10^{-12} \text{ J}$
- B. $3 \times 10^{-10} \text{ J}$
- C. $4 \times 10^{12} \text{ J}$
- D. $2 \times 10^{18} \text{ J}$

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- *36. For the following nuclear reaction, state the type of reaction and determine the energy released.



Given: mass of ${}^2_1\text{H} = 2.014 \text{ u}$

mass of ${}^3_1\text{H} = 3.016 \text{ u}$

mass of ${}^4_2\text{He} = 4.003 \text{ u}$

mass of ${}^1_0\text{n} = 1.009 \text{ u}$

	Type of reaction	Energy released
A.	fusion	0.018 MeV
B.	fusion	16.76 MeV
C.	fission	0.018 MeV
D.	fission	16.76 MeV

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34. Which of these is a nuclear fusion reaction ?

- A. ${}^{235}_{92}\text{U} + \text{n} \rightarrow {}^{144}_{56}\text{Ba} + {}^{90}_{36}\text{Kr} + 2\text{n}$
- B. ${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^4_2\text{He} + \text{n}$
- C. ${}^{14}_7\text{N} + \text{n} \rightarrow {}^{14}_6\text{C} + {}^1_1\text{H}$
- D. ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \alpha$